Ground Penetrating Radar

Our equipment :)

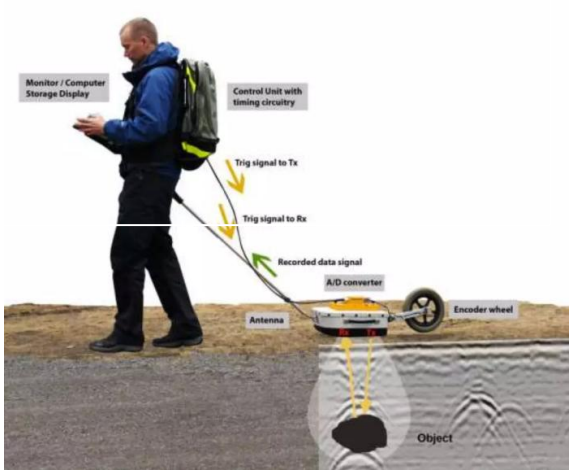
IDS Georadar: The Radar System RIS MF Hi Mod 1A consists mainly of four components:

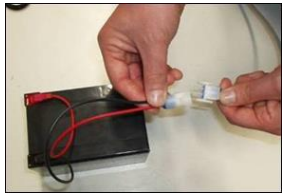
Antenna 200 & 600 MHz

DAD control unit

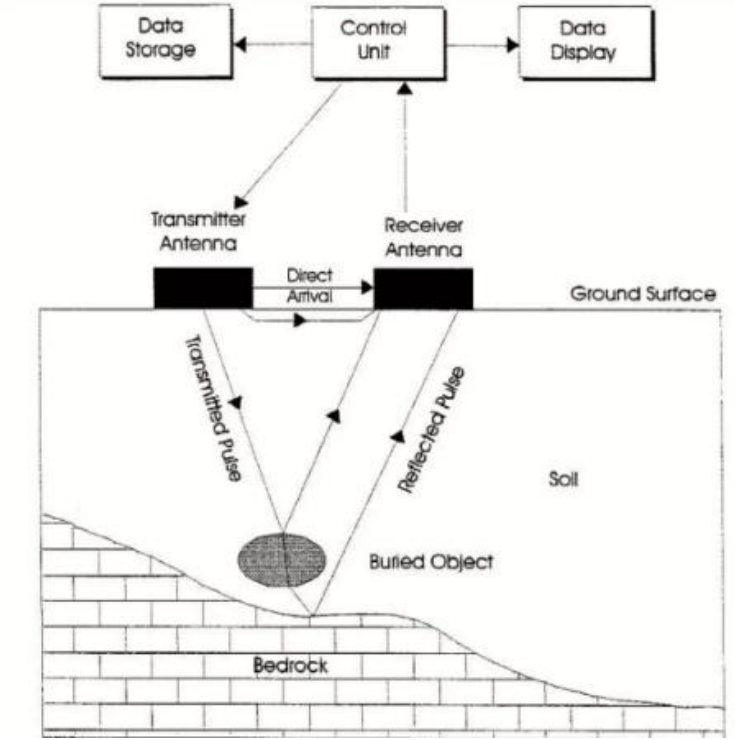
Two batteries

Laptop Panasonic CF -31





Survey and analysis



- GPR uses high frequency radio waves and transmits into the ground.

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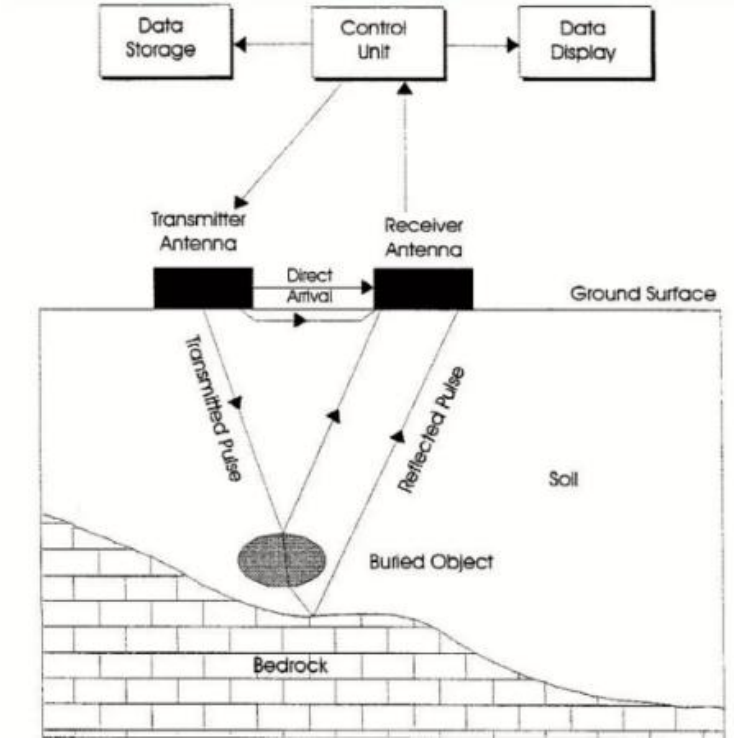

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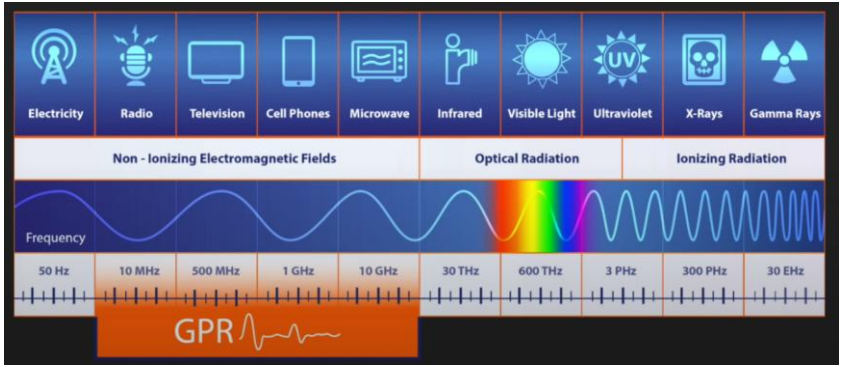
Electromagnetic spectrum



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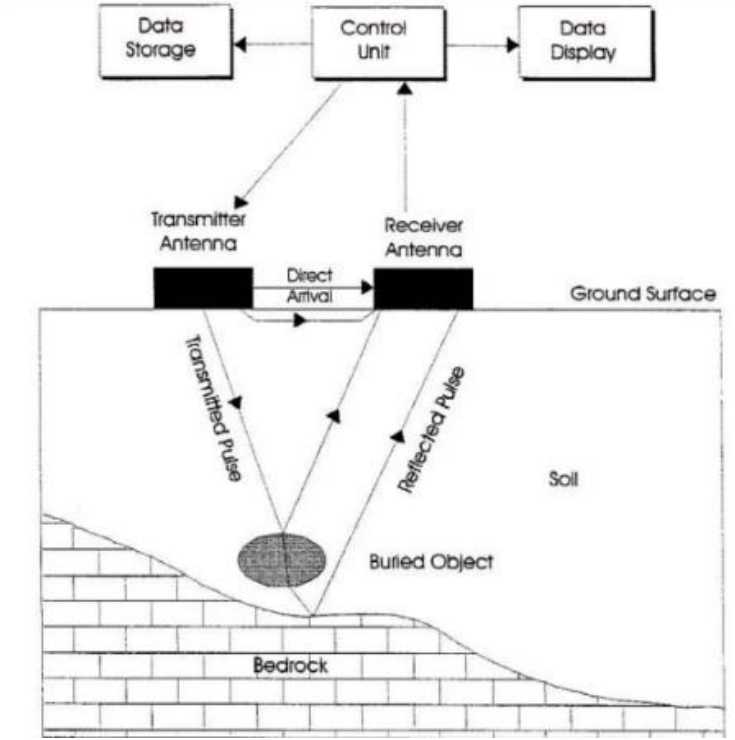

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- When the wave hits a buried object or a boundary with different **dielectric constant,** the receiving antenna records variations in the reflected return signal.

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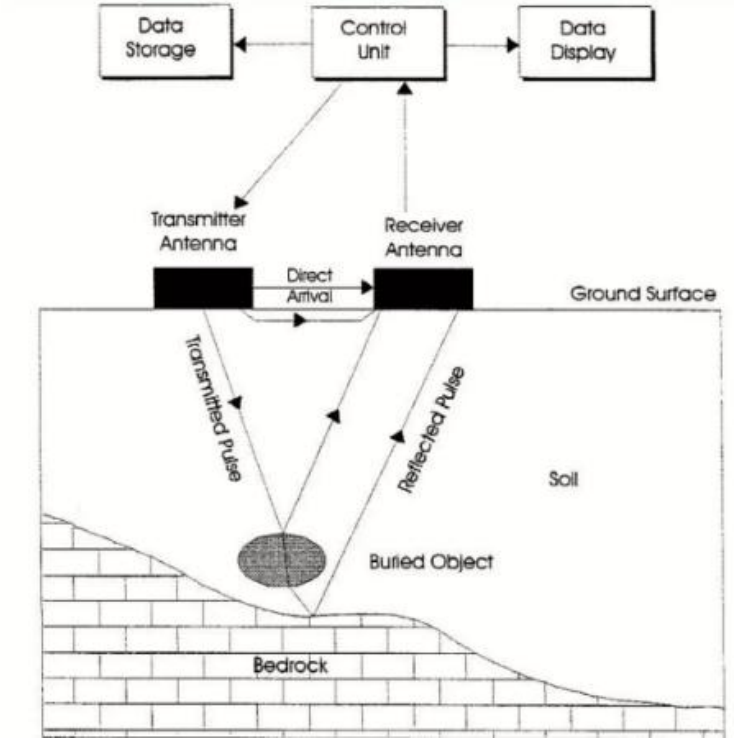

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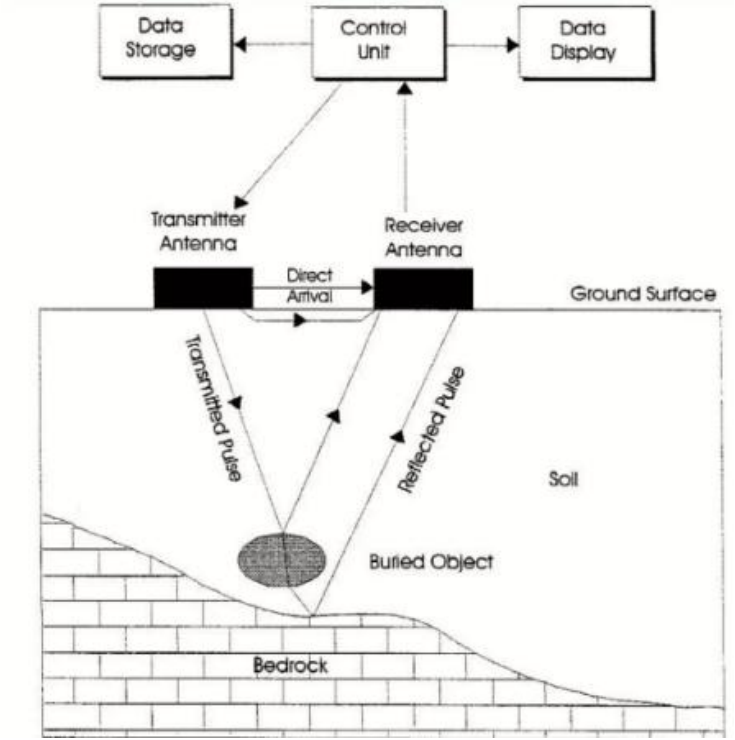

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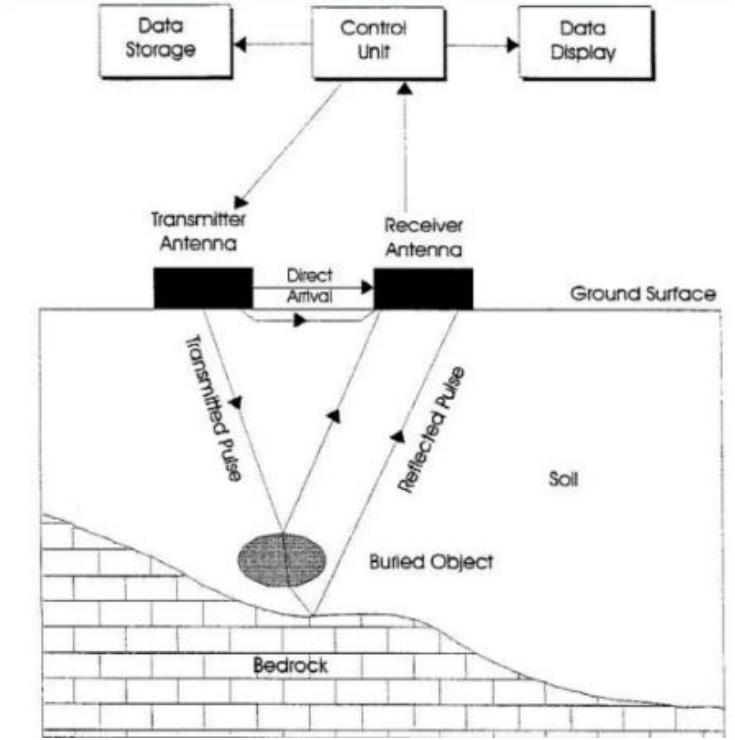
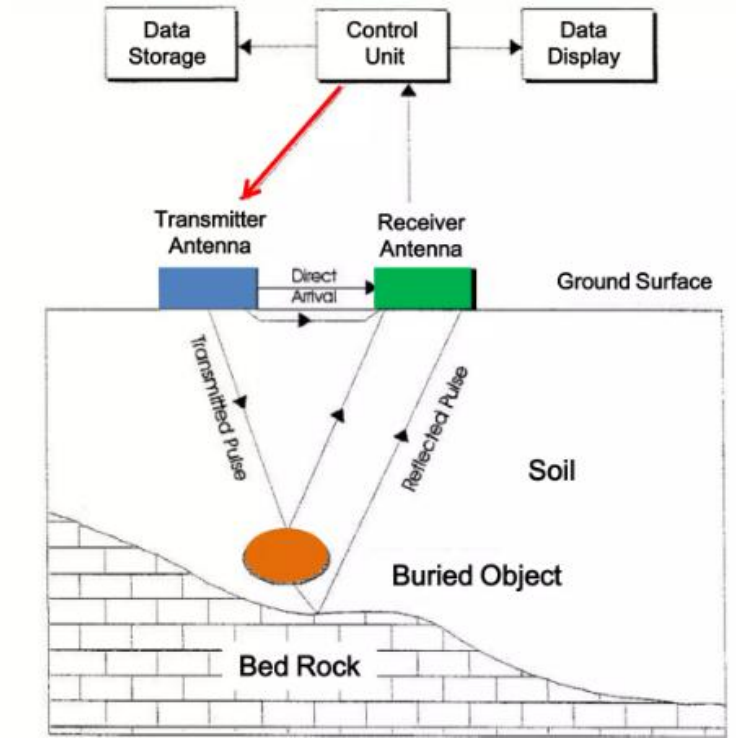
- **Electrical conductivity:** is the ability of a material to support an electrical current due to an applied electrical field. (sigma)
- **Dielectric Permittivity:** is the property that describes the ability of a material to store electric energy by separating opposite polarity charges in space (epsilon, K).

Survey and analysis



- The depth range of GPR is limited by the **electrical conductivity** of the ground, the transmitted **frequency** and the radiated power.
- As **conductivity** increases, the penetration depth decreases.
- Higher **frequencies** do not penetrate as far as lower frequencies, but give better resolution.

Survey and analysis



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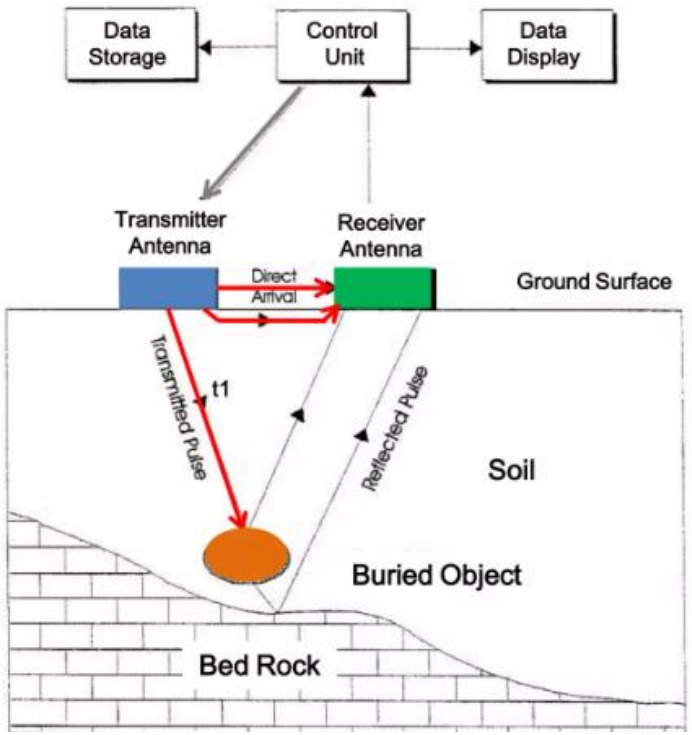

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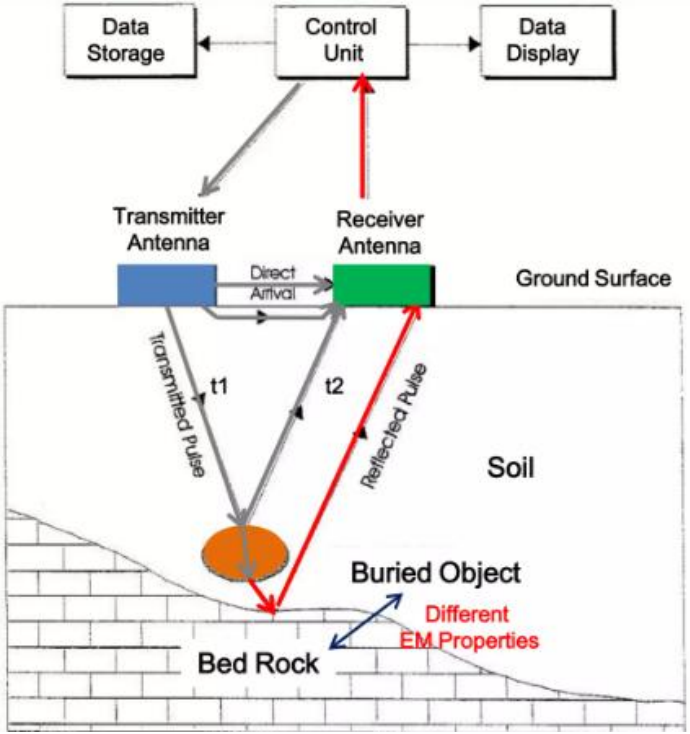
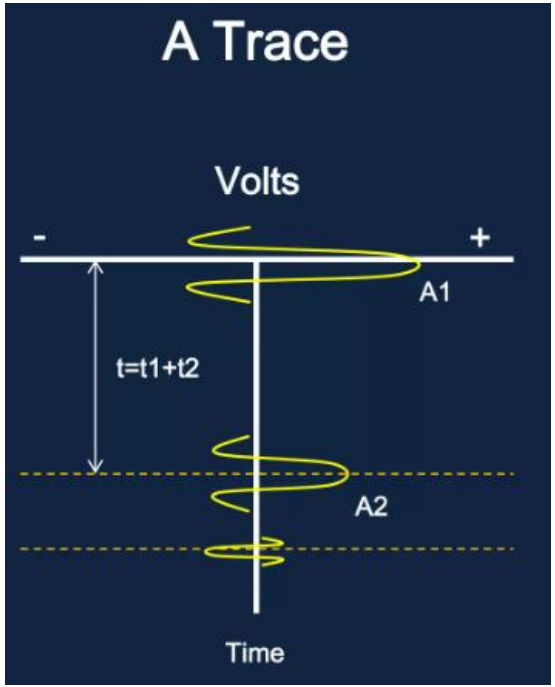


Signal Shape → Amplitude & time



GPR registers the amplitude and the time of the arrival of the signals

Depth $d = \frac{t}{2} \times V_m$

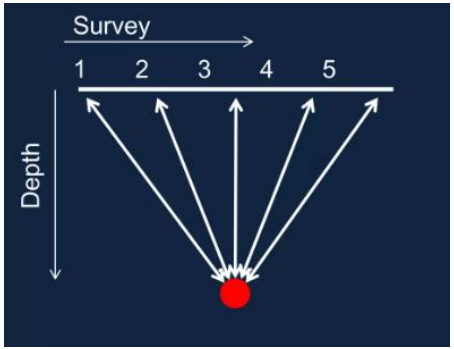
Propagation velocity through the material $V_m = \frac{c}{\sqrt{\varepsilon_r}}$

C propagation velocity in free space

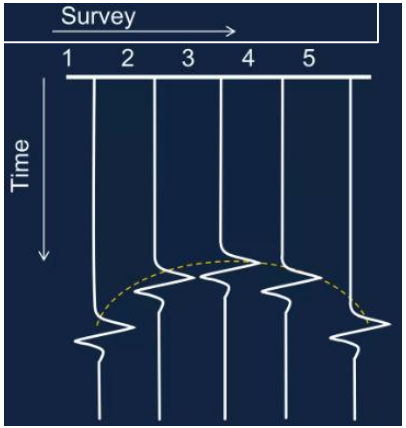
ε_r relative permittivity

Point reflector → signal shape

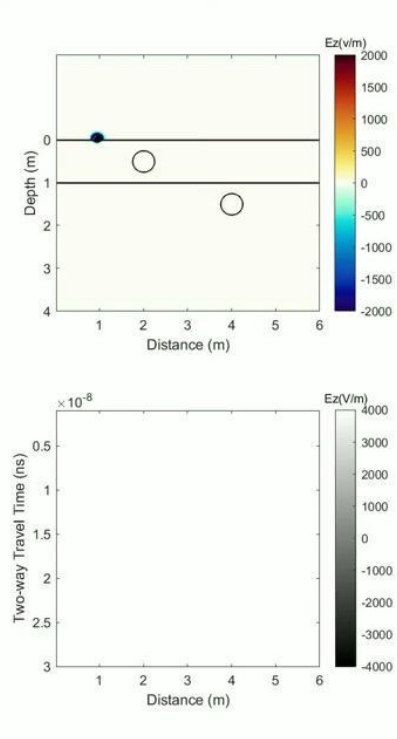
Point Reflector



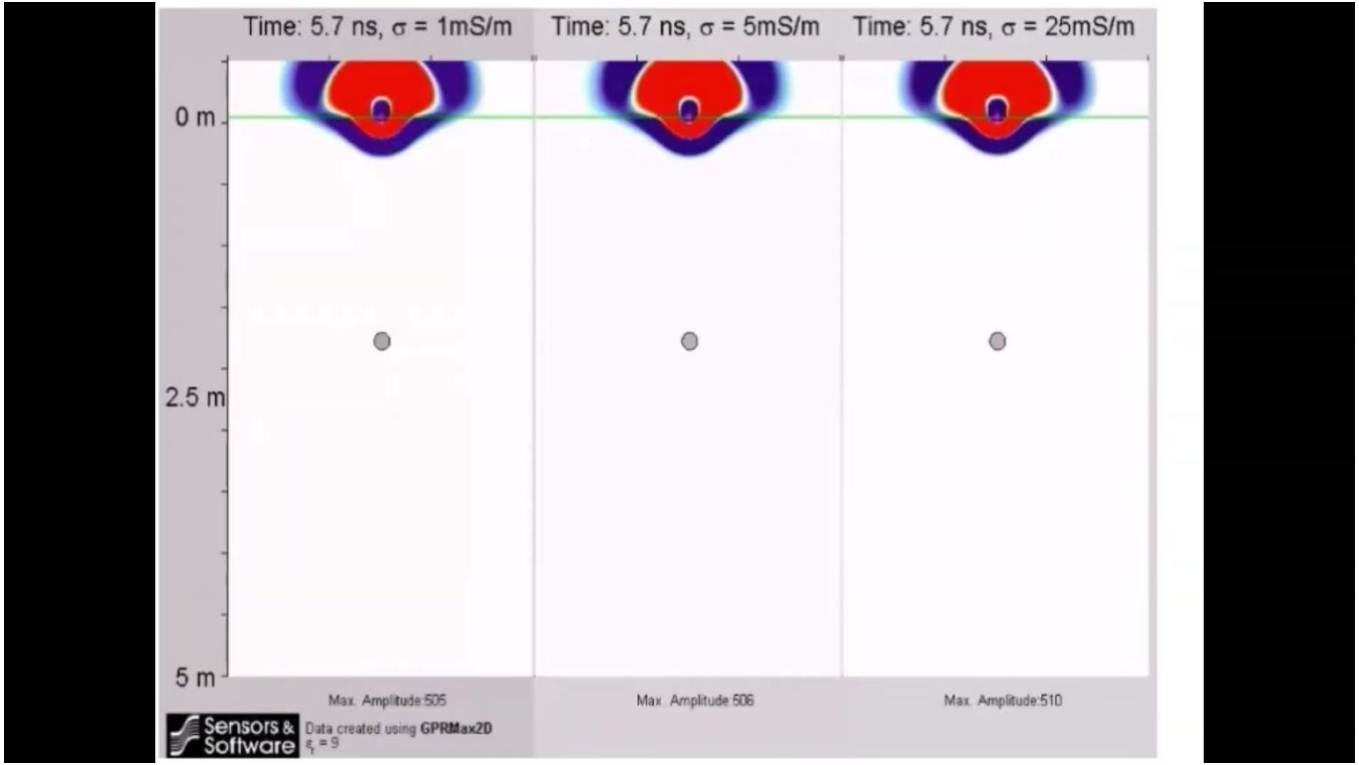
A Series of Traces



Line Scans (BScan, **Radargram**)

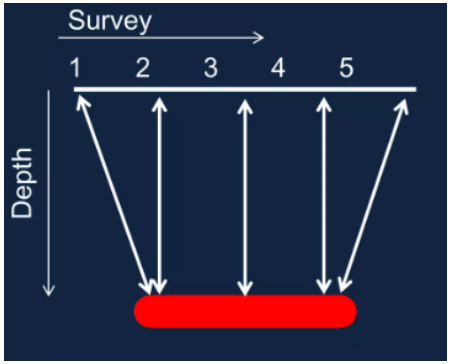


Point reflector → effects when the conductivity of medium changes



Line Reflector → signal shape

horitzontal Reflector



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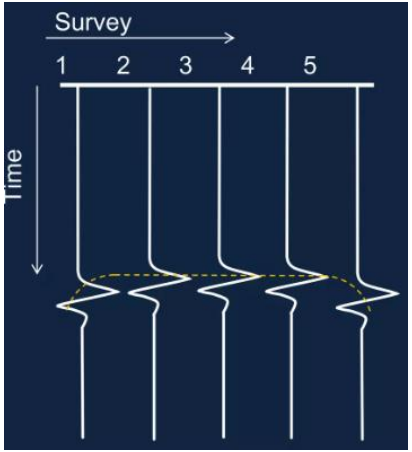
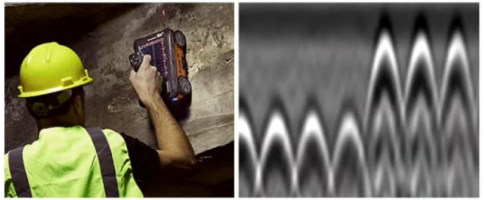
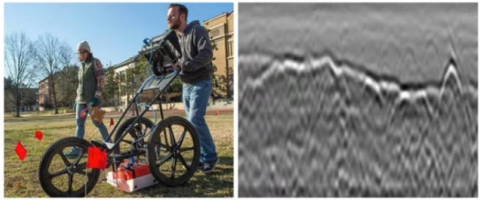


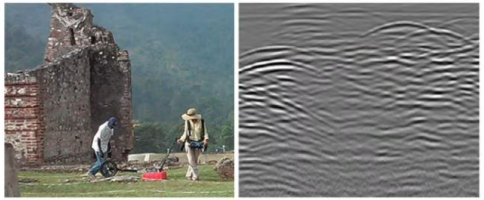
Antenna Frequency



20 in concrete: 2.7 GHz

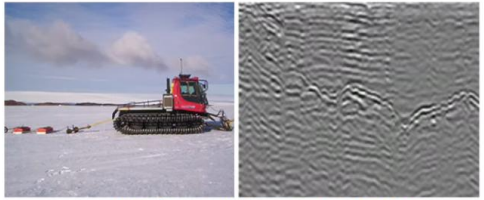


8 ft into the ground: 400 MHz

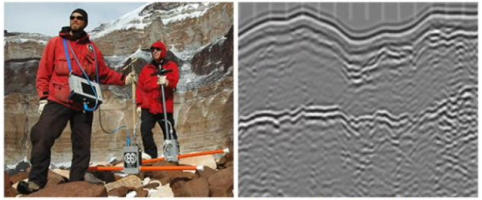


12 ft into the ground: 270 MHz

25 ft into the ground: 200 MHz



100 ft into the ground: 100 MHz



1000 ft in ice: 16 MHz – 80 MHz

Low Frequency

- 12.5 -200 MHz
- Long wavelength
- Deep prospection (up to 150 feet deep)
- Geological and geophysical applications

High Frequency

- 800-2600 MHz
- Short wavelength
- Shallow prospection (as little as 12 inches)
- Concrete scanning and utility locating

Every Material has a Dielectric Constant (K)

Material	K
Air	1
Gasoline	2
Ice	3
Dry Sand	5
Granite	6
Dry Salt	6
Limestone	8
Shale	15
Saturated Sand	25
Silts	30
Clays	40
Distilled Water	80
Fresh Water	80
Sea Water	80
Metal	∞

The Dielectric Constant (K) affects the wave velocity

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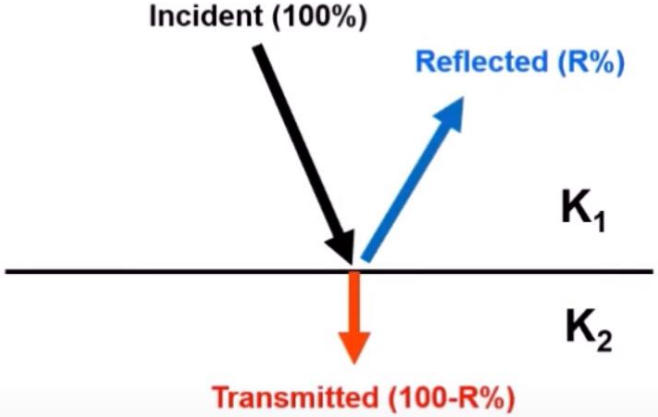
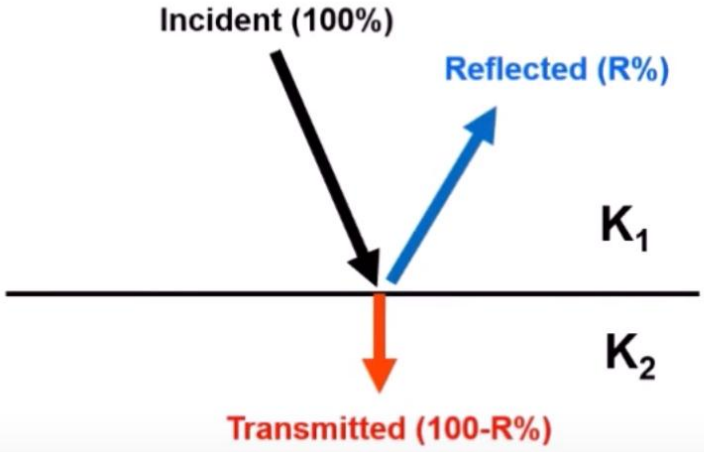
Reflections are created by contrasts



Light showing reflection and transmission

Reflections are created by contrasts



$$R = \frac{\sqrt{K_1} - \sqrt{K_2}}{\sqrt{K_1} + \sqrt{K_2}}$$

Reflection Strength depends on the contrast of materials

Material 1	***K1***	***Material 2***	***K2***	***Reflectivity***
Dry Sand	5	Rock	6	-5%
Wet Sand	25	Rock	6	34%
Dry Sand	5	Wet Sand	25	-38%
Rock	6	Air	1	42%
Water	80	Rock	6	57%
Ice	3	Water	80	67%
Soil	12	Metal	infinito	100%

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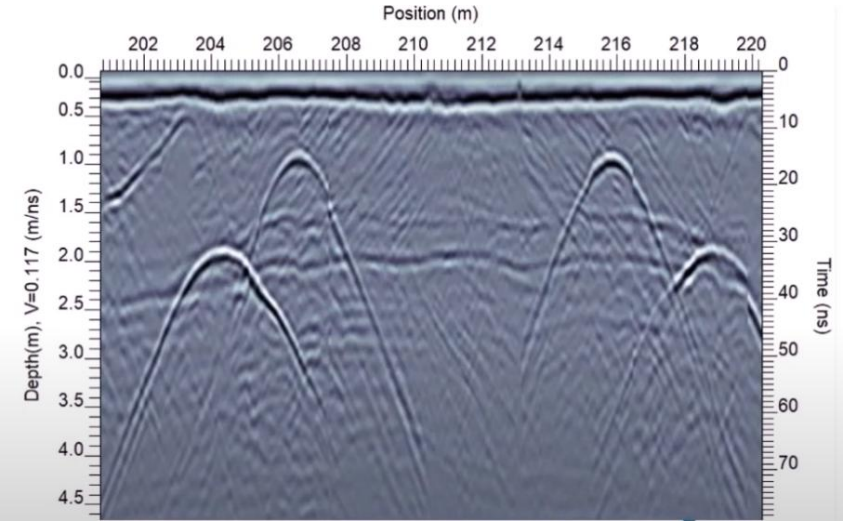
Dry Sand over Rock



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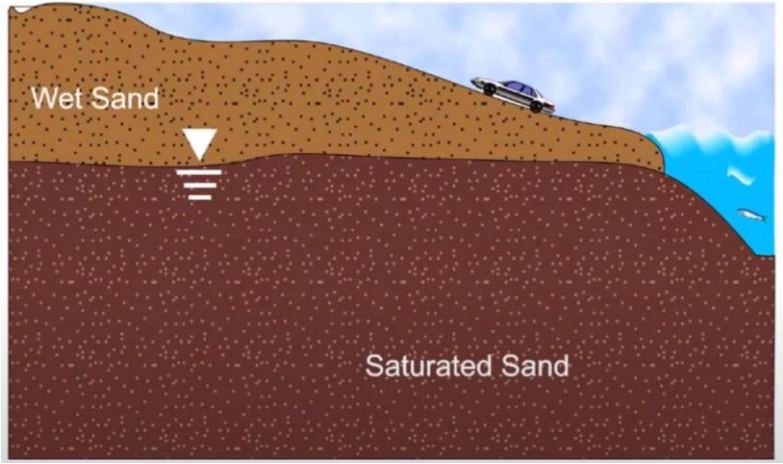
Dry Sand over Wet Sand



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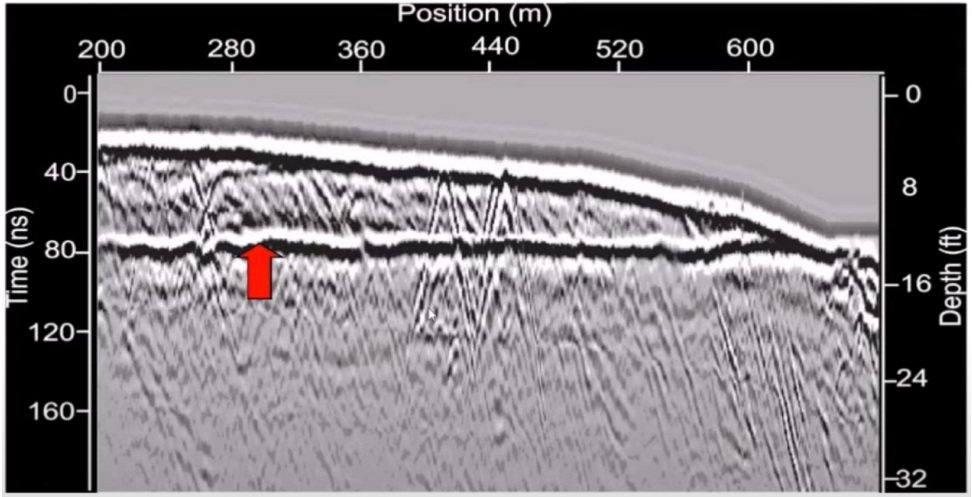

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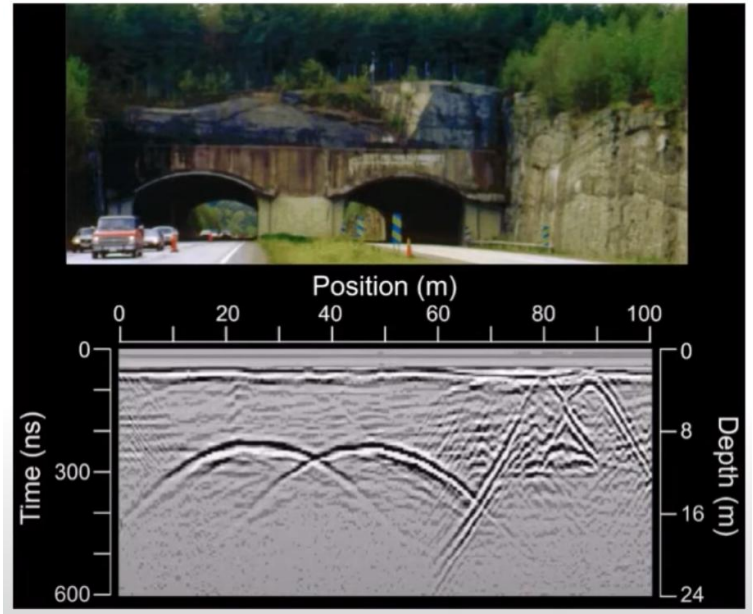
Cavity in Rock



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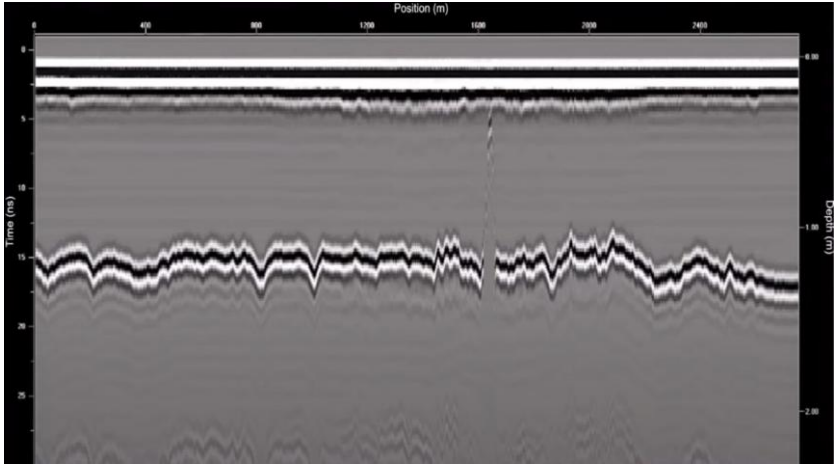
Ice over Water



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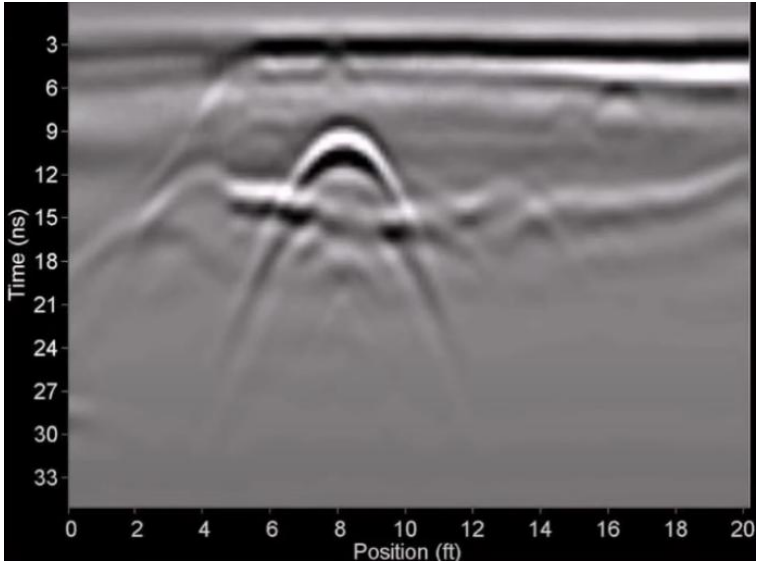
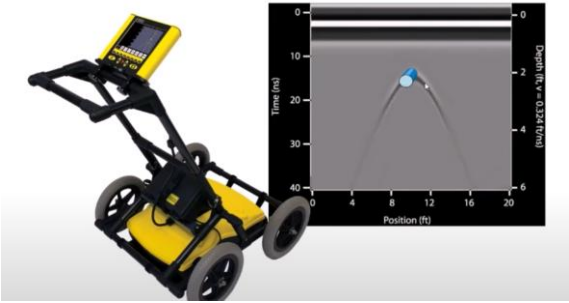
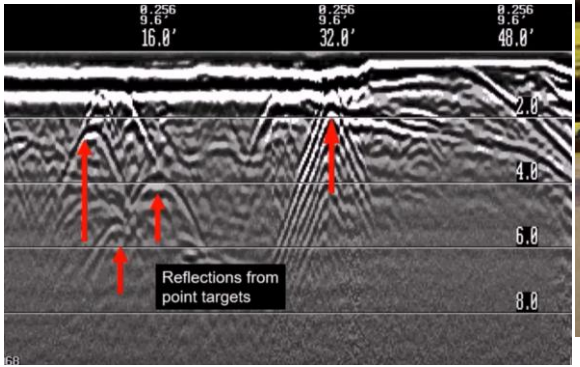
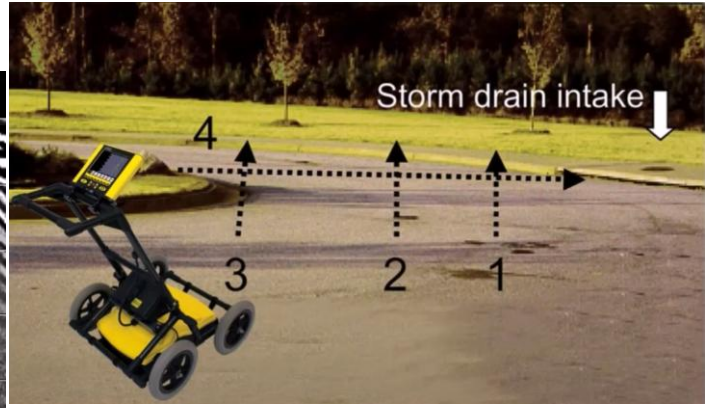
Metal Utility in Soil

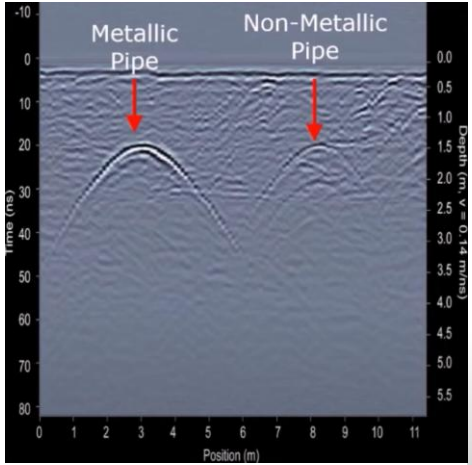


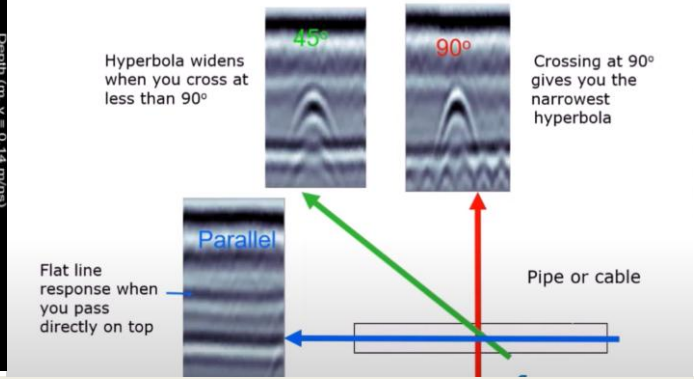
Today → metallic and non metallic pipes

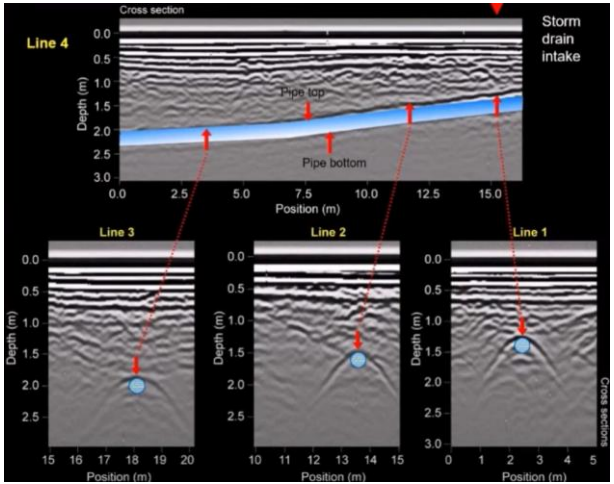






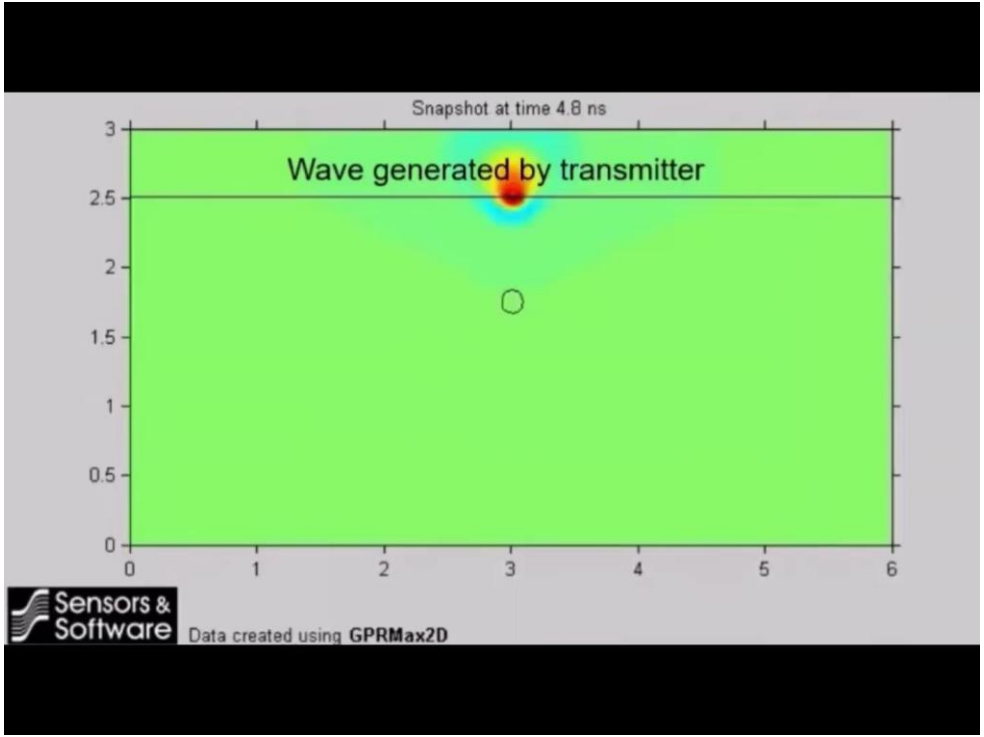






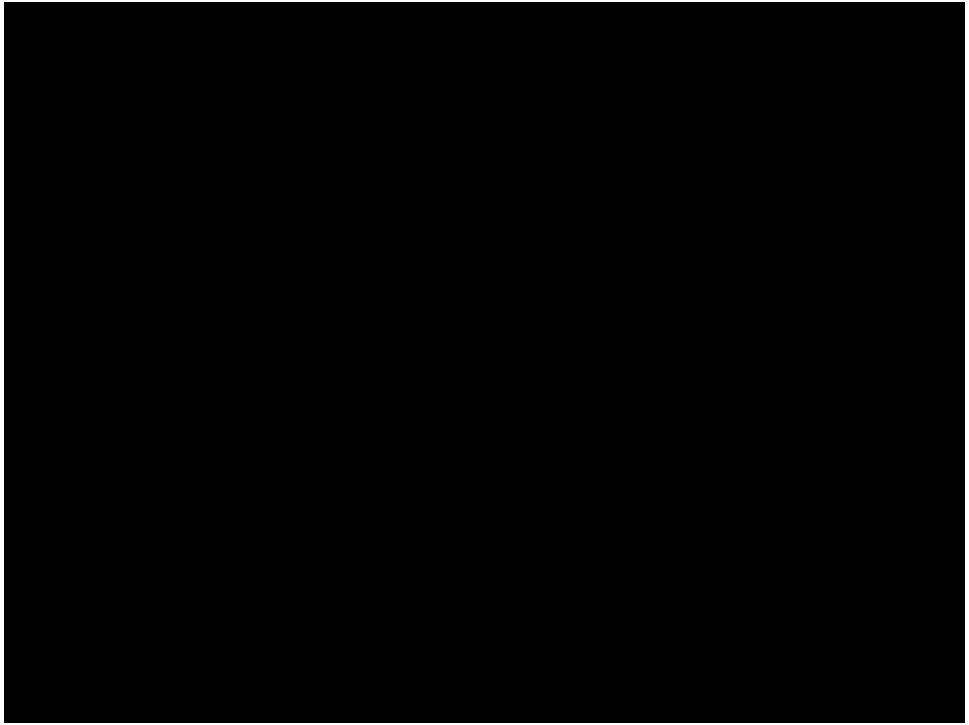
Summary

Computer simulation of GPR signals



Summary

Reflections at boundaries



Summary

GPR is able to detect boundaries between materials with different dielectric permittivity.

→ The greater the contrast, the better the signal reflection.

Electromagnetic waves travel differently depending on the dielectric permittivity of the soil.

→ This difference helps determine the depth of objects using the travel time formula.

$$depth = \frac{c}{\sqrt{\varepsilon_r}} * \frac{time}{2}$$

Soil conductivity affects signal attenuation.

→ The higher the conductivity, the greater the interaction between the signal and the soil, leading to more energy dissipation. This leading to lower penetration depths.

The frequency of the antenna affects depth and resolution.

→ **Low-frequency antennas** penetrate deeper but offer lower resolution.

→ **High-frequency antennas** provide better resolution but have limited penetration depth.

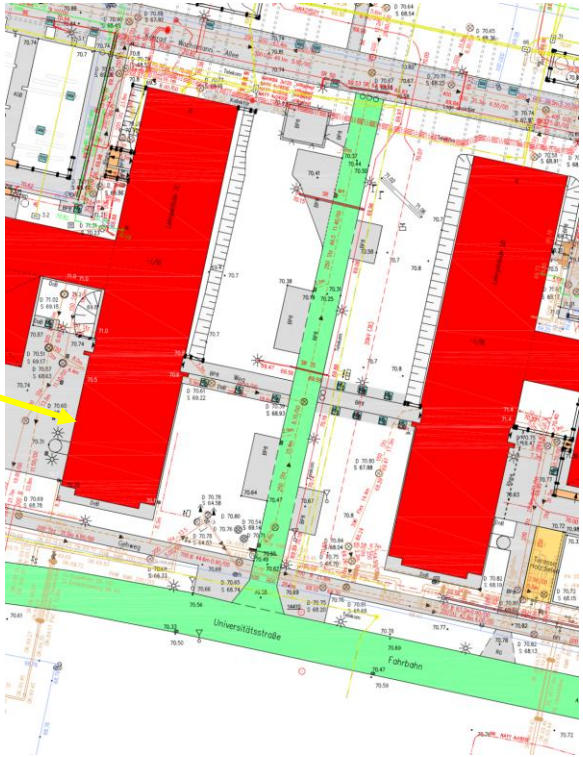
Done! … Time to use the Radar!

Done! … Time to use the Radar!

but before that…

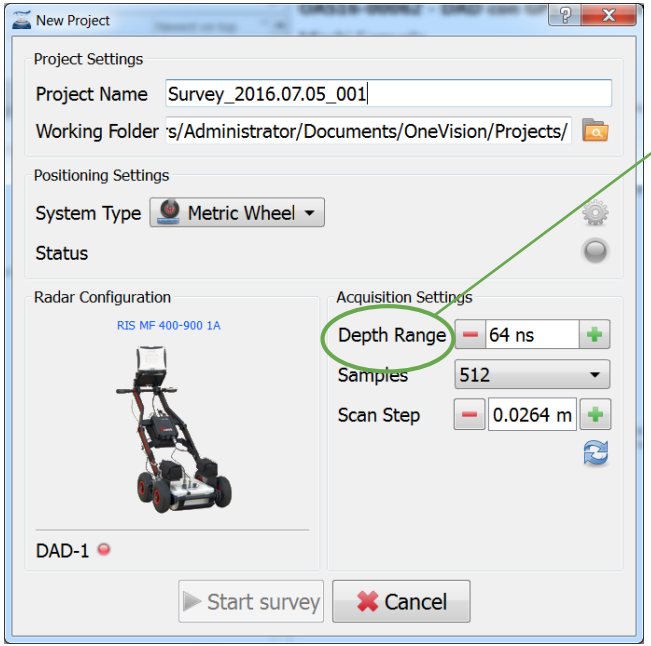
Survey Planning → What are we planning to find?

Utility plan: location of cables, pipes, and similar infrastructure both below and above the ground.



Let's check in we are able to locate utilities like: Storm sewer, Sewer pipe, Power line, Flex line and the differences they show in a radargram

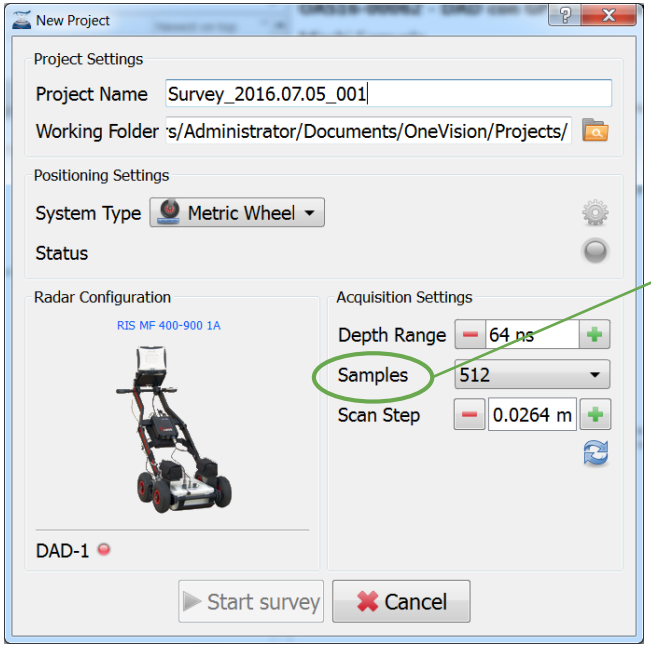
Acquisition software: important parameter setting



$$propagation\ speed = \frac{c}{\sqrt{\varepsilon_r}}$$

$$depth = \frac{velocity * time}{} = \frac{propagation\ speed * range}{2} = \frac{c}{\sqrt{\varepsilon_r}} * \frac{range}{2}$$

Acquisition software: important parameter setting

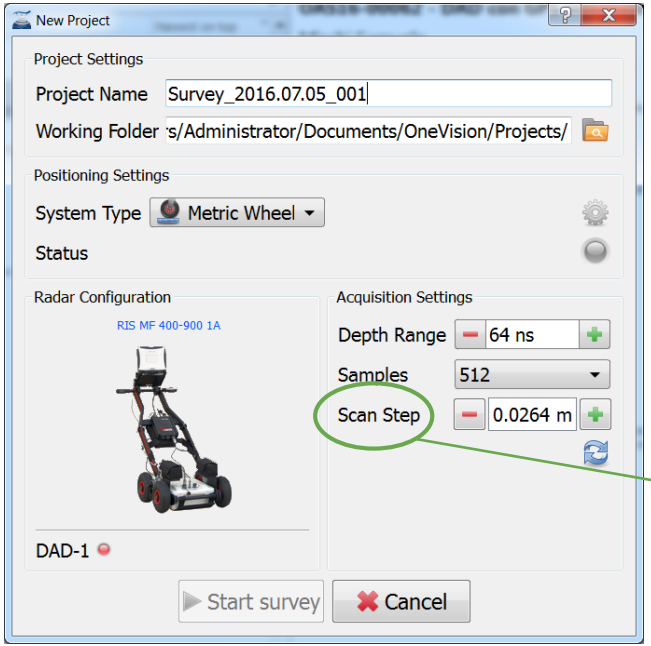


check aliasing problem →

$$samplig\ frequency = \frac{number\ of\ samples}{range} \gg 2 * f_{MAX} = 2 * 600MHz = 1200MHz$$

example: range=70ns, samples=512 provides samplig frequency around 7.30 GHz, which is ok

Acquisition software: important parameter setting



Lateral resolution

Careful! Electromagnetic noise

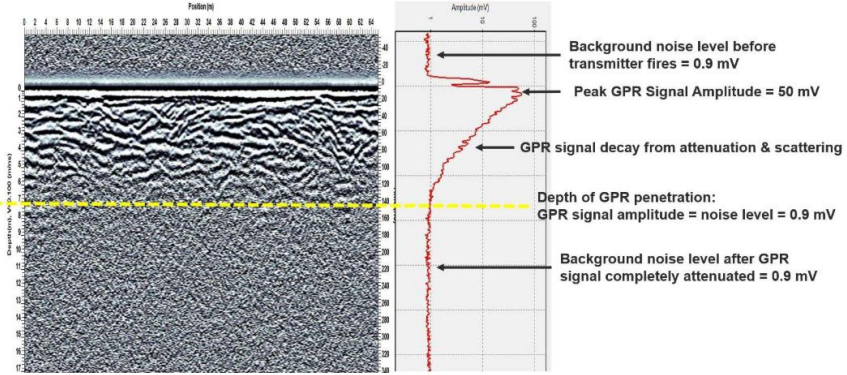


Figure 1
GPR data line (left) and Average Trace Amplitude plot (right) with a typical background noise level of 0.9 mV compared to a peak GPR signal amplitude of 50 mV.